DAY 03 – NETWORK INTERFACES

#THEORY

1. What is a Network Interface Card (NIC)?

A Network Interface Card (NIC) is a hardware component that allows a computer or device to connect to a network. It enables communication and data transfer between devices over a wired or wireless network.

2. What is Ethernet?

Ethernet is a wired networking technology used to connect devices within a Local Area Network (LAN). It allows devices to communicate and exchange data using Ethernet cables.

3. What is RJ-45?

RJ-45 (Registered Jack 45) is a standard connector commonly used for Ethernet networking. It is used to connect network cables to computers, routers, switches, and other networking devices.

4. What does UTP stand for?

UTP stands for Unshielded Twisted Pair. It is a type of networking cable containing pairs of twisted copper wires without additional shielding. Twisting helps reduce electromagnetic interference.

5. Name three common Ethernet cable categories.

Three common Ethernet cable categories are:

- Cat5e (Category 5 Enhanced)
- Cat6 (Category 6)
- Cat6a (Category 6 Augmented)

6. What is a straight-through cable?

A straight-through cable is an Ethernet cable in which both ends follow the same wiring standard (T568A or T568B). It is commonly used to connect different types of networking devices, such as a computer to a switch.

7. What is a crossover cable?

A crossover cable is an Ethernet cable in which the two ends follow different wiring standards (one T568A and the other T568B). It is traditionally used to connect similar networking devices, such as two computers directly.

8. Explain Auto MDI-X.

Auto MDI-X (Automatic Medium-Dependent Interface Crossover) is a networking feature that automatically detects the type of Ethernet cable connection and adjusts the transmit and receive signals accordingly. It eliminates the need to manually use straight-through or crossover cables.

9. Difference between copper and fiber-optic networking?

Copper Networking	**Fiber-Optic Networking**
Uses copper wires to transmit data.	Uses light signals to transmit data.
Generally supports shorter distances.	Supports longer transmission distances.
More affected by electromagnetic interference.	Not affected by electromagnetic interference.
Generally has a lower installation cost.	Generally has a higher installation cost.
Commonly uses RJ-45 connectors.	Uses fiber-optic connectors.

10. What is multimode fiber?

Multimode fiber is a type of optical fiber that allows multiple light paths (modes) to travel through its core. It is commonly used for short-distance communication in buildings, offices, and data centers.

11. What is single-mode fiber?

Single-mode fiber is a type of optical fiber that allows only one light path (mode) to travel through its very small core. It is used for long-distance communication and provides high bandwidth with low signal loss.

12. Two advantages of fiber-optic networking.

1. High Speed: Fiber-optic cables support high data transmission speeds and provide high bandwidth.
2. Long-Distance Communication: They can transmit data over long distances with very little signal loss.

13. Why is understanding network interfaces important for cybersecurity?

Understanding network interfaces is important for cybersecurity because:

1. It helps identify devices connected to a network.
2. It helps detect unauthorized network connections.
3. It helps understand how data enters and leaves a system.

4. It supports network monitoring and troubleshooting.
5. It helps identify possible network vulnerabilities and security threats.

#PRACTICAL IDENTIFICATION

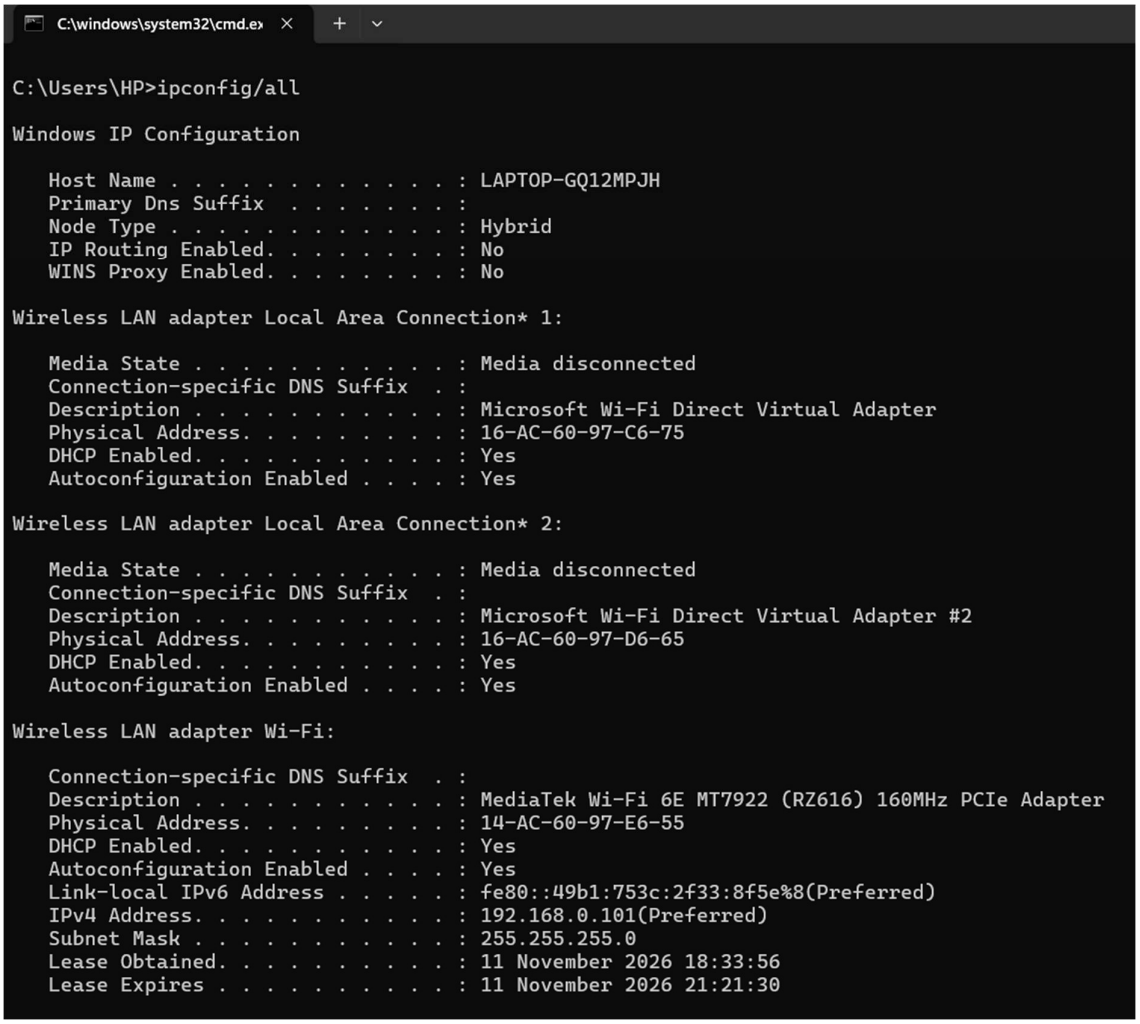
14. Identify your active network adapter.

My active network adapter is Wireless LAN adapter Wi-Fi.

15. Write your IPv4 address, default gateway, and network adapter name.

- Network Adapter Name: Wireless LAN adapter Wi-Fi
- IPv4 Address: 192.168.0.101
- Default Gateway: 192.168.0.1

#SCREENSHOT (3 marks) Attach a screenshot of: ipconfig /all



```
   DHCP Server . . . . . . . . . . . : 192.168.0.1
   DHCPv6 IAID . . . . . . . . . . . : 85240928
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-32-46-78-B6-48-EA-62-93-48-CB
   DNS Servers . . . . . . . . . . . : 192.168.0.1
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek Gaming GbE Family Controller
   Physical Address. . . . . . . . . : 48-EA-62-93-48-CB
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

C:\Users\HP>
```